Project Group -

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Research Objective

Requires data modeling and quantitative research in Transport, Infrastructure & Logistics

To develop a dashboard that helps crowd managers identify peak visitor flows and sudden changes in pedestrian flows at selected sensor locations during SAIL2025.

At which selected locations and times were the highest visitor flows recorded during SAIL2025? How did visitor flows at these locations change throughout the event? Where and when did sudden increases or decreases in pedestrian flows occur? How can these patterns and sudden changes be clearly presented in a dashboard for crowd managers?

Contribution Statement

Be specific. Some of the tasks can be coding (expect everyone to do this), background research, conceptualisation, visualisation, data analysis, data modelling

Anoushka: Conceptualisation of the research objective and research questions; coordination of the project and overall structure of the report. Design the dashboard concept from the crowd manager's perspective. Coding: builds the dashboard layout and integrates the visualisations produced by the team.

Nouschka: Exploratory data analysis of the LVMA dataset. Coding: data inspection and cleaning, including handling missing values, duplicated measurements, incorrect timestamps, inconsistent sensor identifiers and outliers, and filtering the data to the SAIL2025 event period (20—24 August).

Julia: Data integration and modelling. Coding: linked the pedestrian counts to the sensor-location dataset and calculates the indicators (counts per three-minute interval, flow per metre of effective walking width). Created the map-based visualisation of sensor locations.

Lieve: Data analysis focused on peaks and sudden changes. Developes the method for detecting sudden increases and decreases. Coding: implements peak flow per sensor, time of peak flow, and change detection.

Laure: Background research on crowd management and pedestrian flow indicators at large-scale events, used to interpret flow levels and choose meaningful thresholds. Coding: comparison of flow patterns across locations, event days and periods of the day, including the time-series graphs.

Data Used

The main datasets used in this study are: SAIL2025 LVMA_3min_20August25August2025_flow sensor-location

We will use pedestrian counts from the LVMA sensor dataset, recorded at three-minute intervals during SAIL2025. These measurements will be combined with the sensorlocation.xlsx dataset, which provides the geographical locations of the sensors and their effective pedestrian widths.

This combination allows us to examine absolute pedestrian flows as well as pedestrian flow per metre of effective walking width. We will use these indicators to compare crowd conditions between selected locations, identify peak periods and detect sudden changes over time.

Where a sensor records pedestrian movements in two directions, we will first examine the directional measurements separately. Depending on the research question and sensor configuration, these flows may subsequently be combined to represent the total pedestrian flow at that location.

Data Pipeline

First, we will load the LVMA pedestrian counts and sensor location data in Python. We will conduct a preliminary analysis to inspect the available sensors, measurement period and data quality. Next, we will clean the measurements, select the SAIL2025 event period and link the counts to their sensor locations and effective widths. We will then calculate and compare visitor flows, identify peaks and sudden changes over time, and present the results in dashboard visualisations.

First, the LVMA pedestrian count data and sensor-location data will be imported into Python. An exploratory analysis will be performed to understand the available sensors, variables, measurement period, measurement frequency and overall data quality.

The following analysis pipeline will then be used:

1. Data inspection and cleaning Check for missing values, duplicated measurements, incorrect timestamps, inconsistent sensor identifiers and potential outliers.

2. Temporal selection Filter the measurements to the SAIL2025 event period from 20 to 24 August 2025.

3. Data integration Link pedestrian measurements to the corresponding sensor locations and effective pedestrian widths.

4. Indicator calculation Calculate indicators such as:

- pedestrian count per three-minute interval;
- pedestrian flow per metre of effective walking width;
- peak pedestrian flow per sensor;
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time of peak flow; • changes in pedestrian flow between consecutive measurements.

5. Peak and change detection Identify the locations and time periods with the highest visitor flows. Sudden increases and decreases will be analysed by examining changes between consecutive time intervals and deviations from typical flow patterns at each sensor.

6. Comparison across locations and time Compare flow patterns between selected sensor locations, event days and different periods of the day.

7. Dashboard visualisation Present the results using clear visualisations such as timeseries graphs, peak-flow indicators, maps and location comparisons. The dashboard will be designed to allow crowd managers to quickly identify locations and periods associated with high or rapidly changing visitor flows.

Geographical Scale The geographical scope of the study is the SAIL2025 event area in Amsterdam. The analysis will focus on selected LVMA pedestrian sensor locations within and around the main SAIL event area.

Temporal Scale The analysis covers the main SAIL2025 event period from 20 August to
24 August 2025. The available dataset contains measurements from 20 to 25 August 2025, but measurements outside the main event period will only be used if they are relevant as a reference or for data validation.

Intended Output The main output of the project will be an interactive crowd-monitoring dashboard. The dashboard will show how pedestrian flows varied across selected locations and over time during SAIL2025. It should enable crowd managers to quickly identify peak crowd periods, locations experiencing relatively high pedestrian pressure and sudden changes in visitor flows.

The analysis will therefore provide both a retrospective assessment of crowd patterns during SAIL2025 and a dashboard structure that could potentially support crowd monitoring during similar large-scale events.